

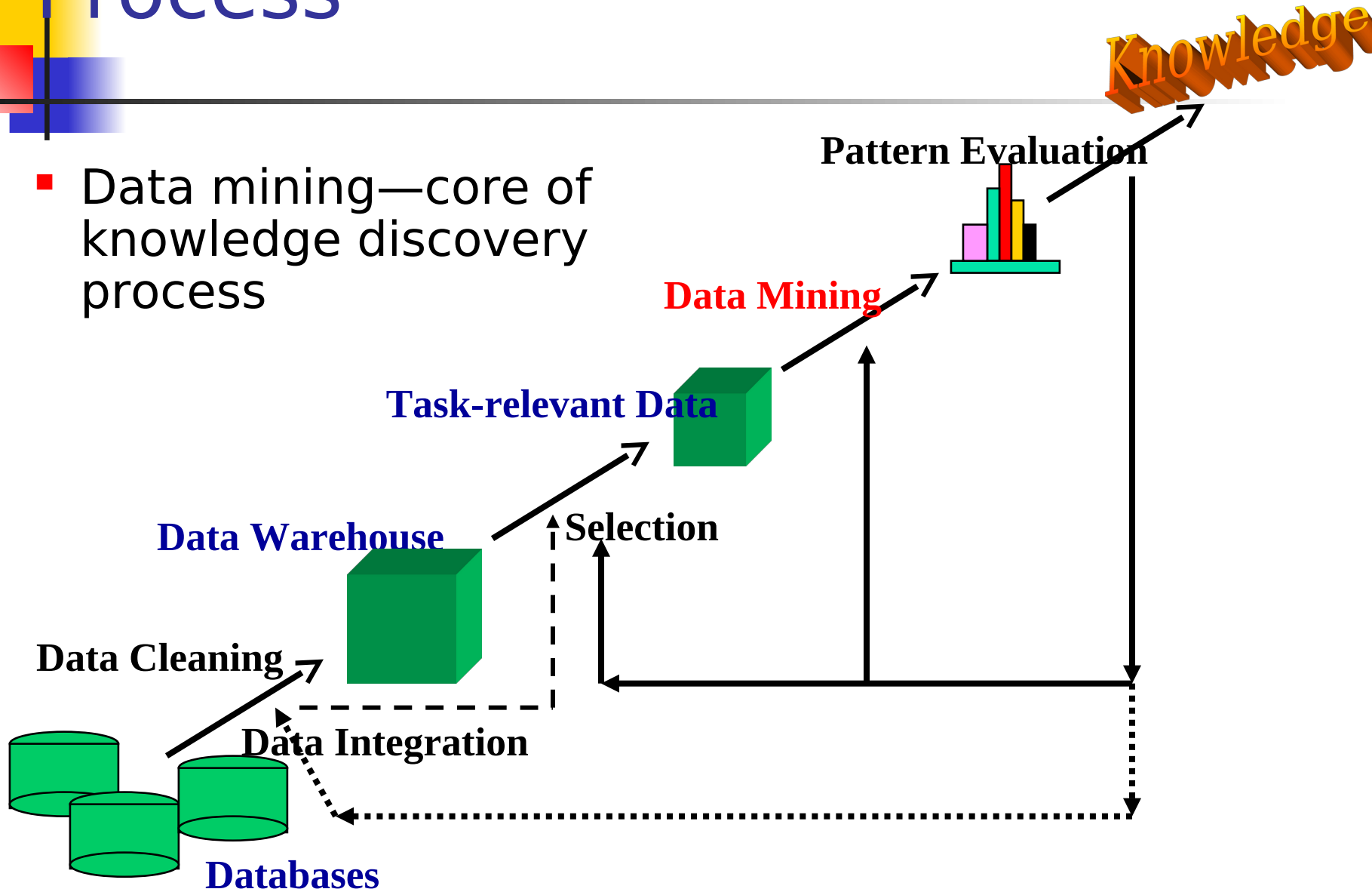
# Introduction to Data Mining

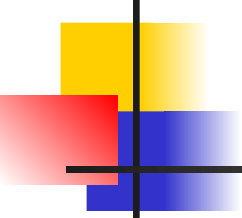
---

Dr. Bernard Chen  
Troy University

# Knowledge Discovery (KDD) Process

- Data mining—core of knowledge discovery process

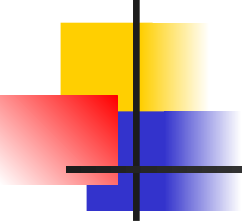




# Knowledge Process

---

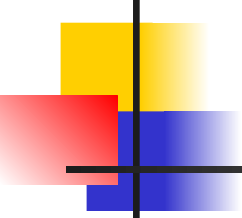
1. **Data cleaning** – to remove noise and inconsistent data
2. **Data integration** – to combine multiple source
3. **Data selection** – to retrieve relevant data for analysis
4. **Data transformation** – to transform data into appropriate form for data mining
5. **Data mining**
6. **Evaluation**
7. **Knowledge presentation**



# Knowledge Process

---

- Step 1 to 4 are different forms of data **preprocessing**
- Although data mining is only one step in the entire process, it is an essential one since it uncovers hidden patterns for evaluation

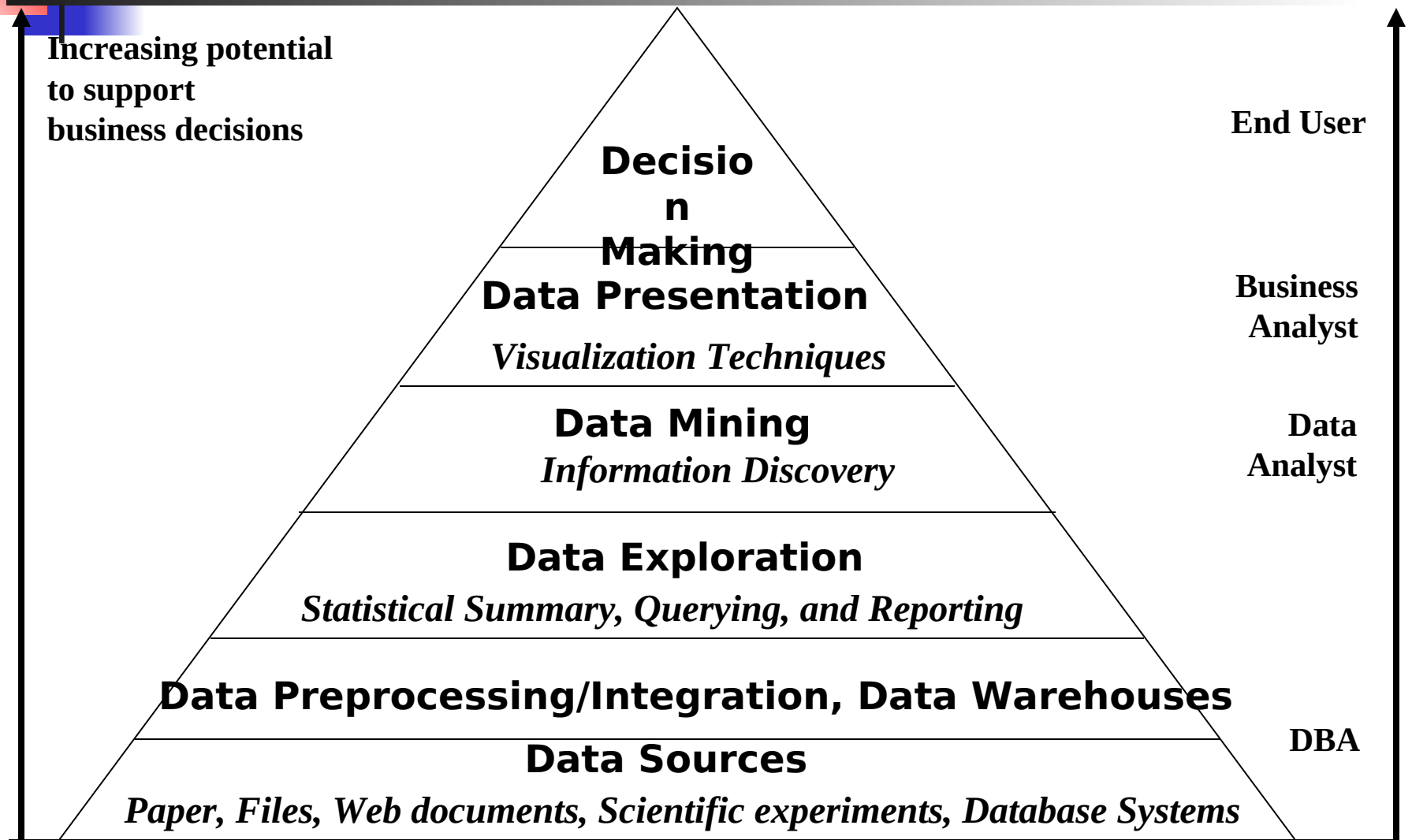


# Knowledge Process

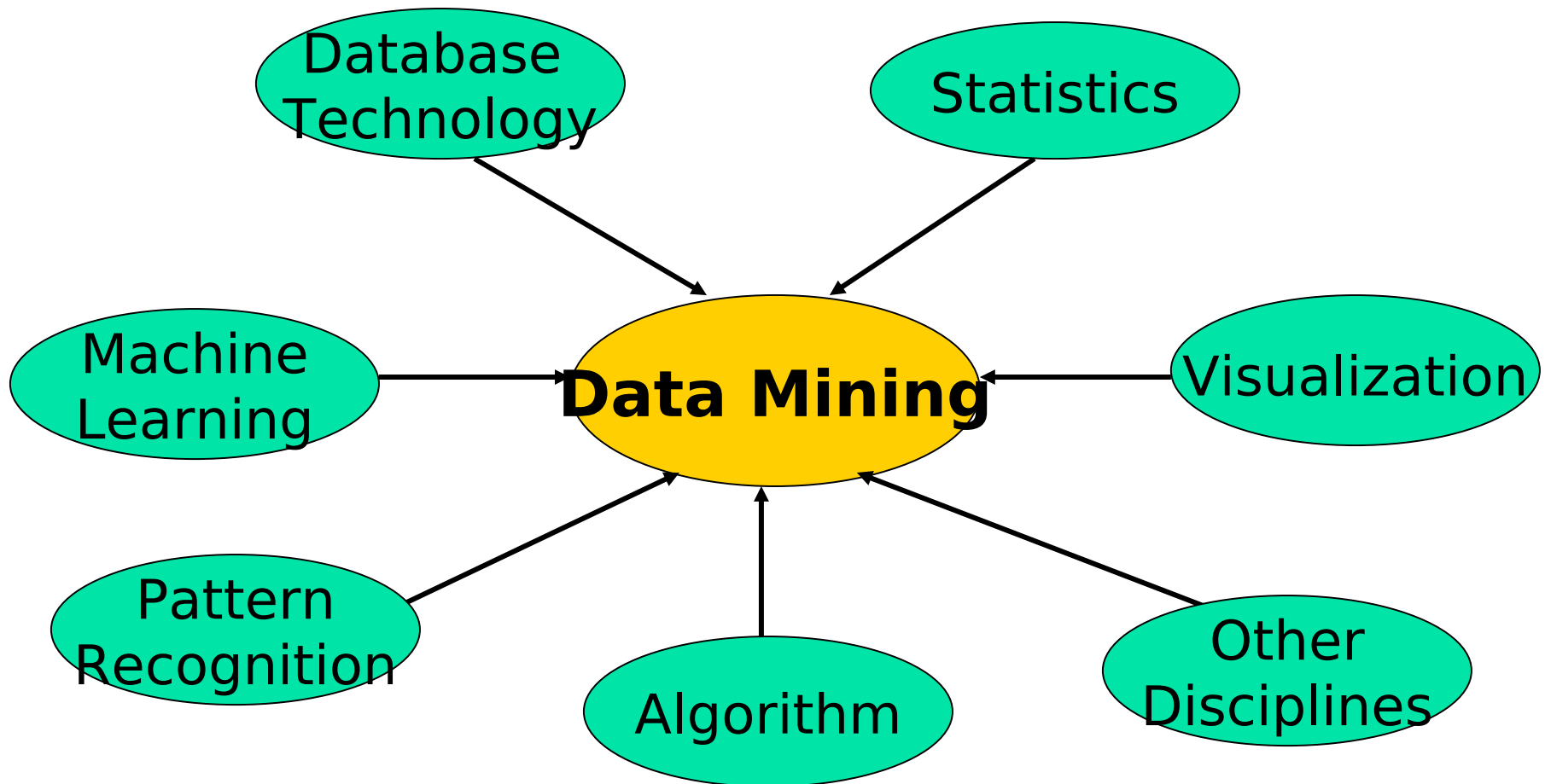
---

- Based on this view, the architecture of a typical data mining system may have the following major components:
  - Database, data warehouse, world wide web, or other information repository
  - Database or data warehouse server
  - Data mining engine
  - Pattern evaluation model
  - User interface

# Data Mining and Business Intelligence

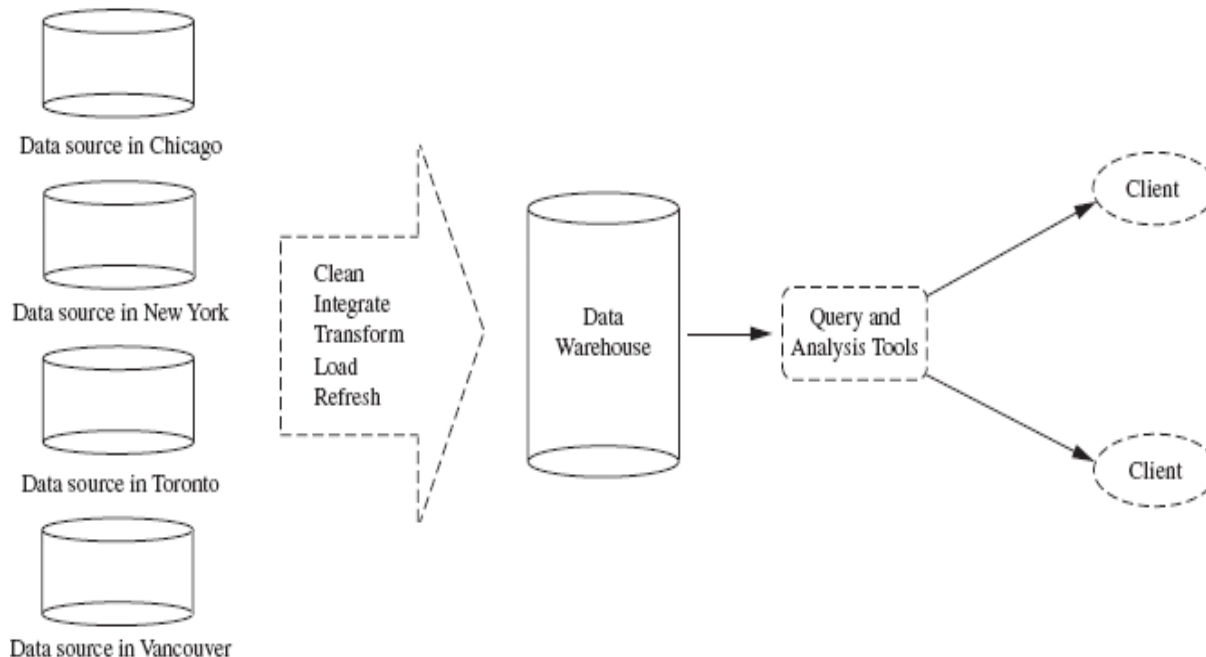


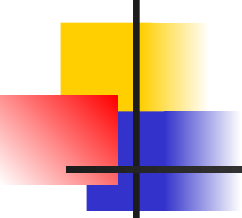
# Data Mining: Confluence of Multiple Disciplines



# Data Mining – on what kind of data?

- Data Warehouses

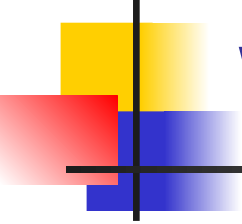




# Data Mining – on what kind of data?

---

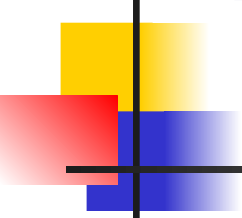
- Transactional Databases
- Advanced data and information systems
  - Object-oriented database
  - Temporal DB, Sequence DB and Time serious DB
  - Spatial DB
  - Text DB and Multimedia DB
  - ... and WWW



# What kind of patterns can we mined?

---

- In general, data mining tasks can be classified into two categories: **descriptive** and **predictive**
  - Descriptive mining tasks characterize the general properties of the data in database
  - Predictive mining tasks performs inference on the current data in order to make predictions



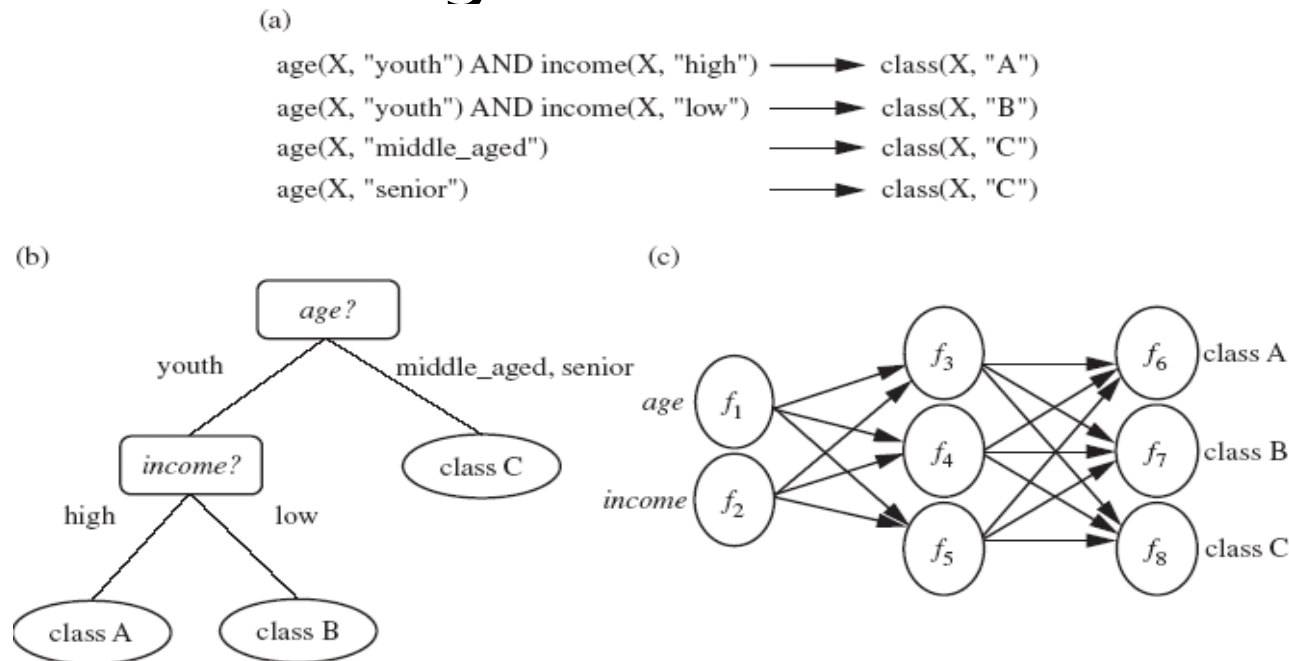
# Mining frequent patterns, Associations, and Correlations (Ch4)

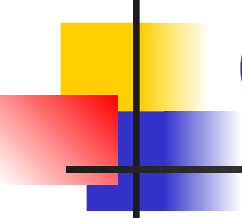
---

- **Frequent patterns** are patterns that occur frequently in data
- Association analysis:
  - Example: `buys(X,"computer") => buys(X,"software")` [support = 1%, confidence = 50%]

# Classification and Prediction (Ch 5)

- **Classification** is the process of finding a MODEL that describes and distinguish data classes or

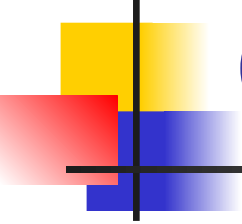




# Cluster analysis (Ch 6)

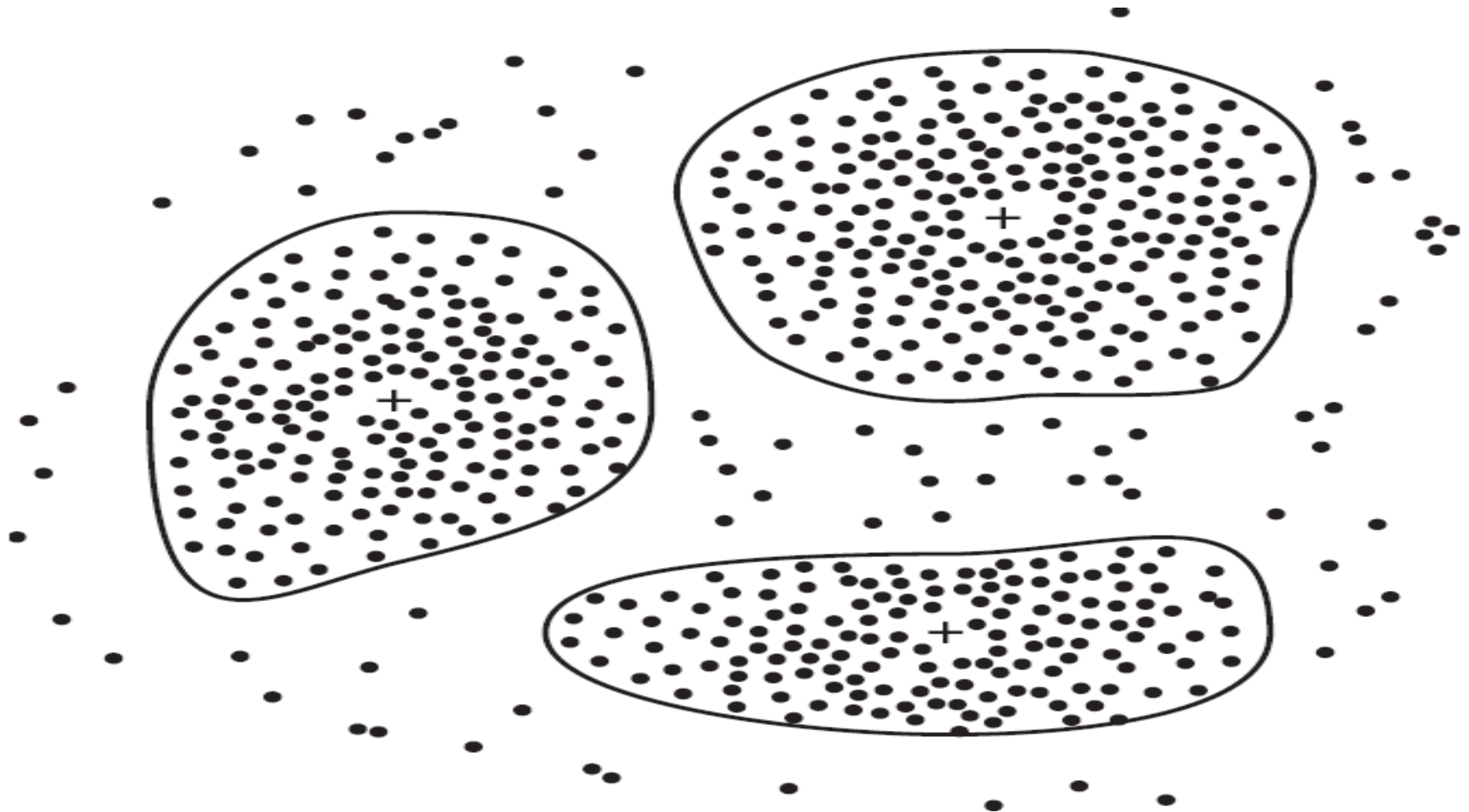
---

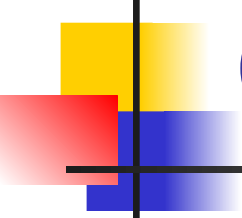
- In general, the class labels are not present in the training data simply they are not known to begin with
- The objects are clustered or grouped based on the principle of *maximizing the intra-cluster similarity* and *minimizing the inter-cluster similarity*



# Cluster analysis

---





# Outlier Analysis (Ch 7)

---

- Most data mining methods discard outliers as noise or exceptions.
- However, in some application such as fraud detection, the rare event can be more interesting than regularly occurring ones